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1 import matplotlib.pyplot as plt
2 import numpy as np
3 import pandas as pd
4 from scipy.optimize import curve_fit
5 from pathlib import Path
6
7 BASE_DIR = Path(__file__).parent
8
9 DATA_PATH = BASE_DIR / "FP66-quant" / "2026-07-20"
10 DATA_KORR_PATH = DATA_PATH / "KorrelationsMessungen"
11 DATA_RATEN_PATH = DATA_PATH / "RatenMessungen"
12 DATA_BIT_PATH = DATA_PATH / "BitMessungen"
13
14 IMAGE_PATH = BASE_DIR / "Images"
15 EINZELZ_PATH = IMAGE_PATH / 'Einzelzählraten'
16 KOINZIDENZ_PATH = IMAGE_PATH / 'Koinzidenz'
17
18 ### ---- Utility Functions ---- ###
19 #
20 ROUND_TO = 3
21 def get_polaristaoinsgrad(df, key):
22     N_max = np.max(df[key])
23     N_min = np.min(df[key])
24     return (N_max - N_min) / (N_max + N_min)
25
26 def get_correlation(df: pd.DataFrame):
27     deg_alice = np.array([df["AngleAlice(deg)"]])
28     deg_bob = np.array([df["AngleBob(deg)"]])
29     n_AOB0 = np.array([df["AOB0"]])
30     n_A1B0 = np.array([df["A1B0"]])
31     n_AOB1 = np.array([df["AOB1"]])
32     n_A1B1 = np.array([df["A1B1"]])
33
34
35     corr = (n_AOB0 + n_A1B1 - n_A1B0 - n_AOB1) / (n_AOB0 + n_A1B1 + n_A1B0 + n_AOB1)
36     s_par = corr[0][0] - corr[0][1] + corr[0][2] + corr[0][3]
37
38     A = n_AOB0 + n_A1B1
39     B = n_A1B0 + n_AOB1
40     N = n_AOB0 + n_A1B1 + n_A1B0 + n_AOB1
41     sigma_E = 2 * np.sqrt(A * B / N**3)
42
43     sigma_s = np.sqrt(sum(sigma_E[0][i]**2 for i in range(4)))
44     return ((np.round(corr, ROUND_TO), np.round(sigma_E, ROUND_TO)),
45             (np.round(s_par, ROUND_TO), np.round(sigma_s, ROUND_TO)))
46
47
48 ### ---- Curve fitting Functions ---- ###
49 def func_sin_fitt(x, a, b, c, d):
50     return a * np.sin(b * (x + c)) + d
51
52 def estimate_sin_p0(x, y, n_points=None):
53     """Estimate [a, b, c, d] starting values for a*sin(b*(x+c))+d
54     from the max/min within the first n_points of (x, y)."""
55     x = np.asarray(x)
56     y = np.asarray(y)
57     if n_points is not None:
58         x, y = x[:n_points], y[:n_points]
59
60     idx_max = np.argmax(y)
61     idx_min = np.argmin(y)
62     x_max, x_min = x[idx_max], x[idx_min]
63     y_max, y_min = y[idx_max], y[idx_min]
64
65     print(x_max, x_min, y_max, y_min)
66
67     a0 = (y_max - y_min) / 2
68     b0 = np.pi / np.abs((x_max - x_min))
69     c0 = -(x_max + x_min) / 2 # note the sign fix

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70     d0 = (y_max + y_min) / 2
71
72     return [a0, b0, c0, d0]
73
74
75 err_DEGREE = 0.5
76
77 columns = [
78     'AngleAlice(deg)',
79     'AngleBob(deg)',
80     'AOB0',
81     'A1B0',
82     'AOB1',
83     'A1B1',
84 ]
85
86 data_bob = pd.read_csv(
87     filepath_or_buffer = DATA_KORR_PATH / "2026-07-20_17-03-32_Verschraenkung_Bob.txt",
88     sep = ';',
89     comment = '#',
90     header = None,
91     names = columns
92 )
93 # data_bob.sort_values('AngleBob(deg)', inplace=True, ignore_index=True)
94 data_alice = pd.read_csv(
95     filepath_or_buffer = DATA_KORR_PATH / "2026-07-20_17-03-38_Verschraenkung_Alice.txt",
96     sep = ';',
97     comment = '#',
98     header = None,
99     names = columns
100 )
101 # data_alice.sort_values('AngleAlice(deg)', inplace=True, ignore_index=True)
102 # print(data)
103
104 s_corr_vals = (get_correlation(data_bob))
105 print(
106     f"S Value und Zählraten Bob\n",
107     f" AB   {s_corr_vals[0][0][0][0]} +/- {s_corr_vals[0][1][0][0]} \n",
108     f" A'B   {s_corr_vals[0][0][0][1]} +/- {s_corr_vals[0][1][0][1]} \n",
109     f" AB'   {s_corr_vals[0][0][0][2]} +/- {s_corr_vals[0][1][0][2]} \n",
110     f" A'B'  {s_corr_vals[0][0][0][3]} +/- {s_corr_vals[0][1][0][3]} \n",
111     f" S-Value {s_corr_vals[1][0]} +/- {s_corr_vals[1][1]} \n",
112 )
113 s_corr_vals = (get_correlation(data_alice))
114 print(
115     f"S Value und Zählraten Alice\n",
116     f" AB   {s_corr_vals[0][0][0][0]} +/- {s_corr_vals[0][1][0][0]} \n",
117     f" A'B   {s_corr_vals[0][0][0][1]} +/- {s_corr_vals[0][1][0][1]} \n",
118     f" AB'   {s_corr_vals[0][0][0][2]} +/- {s_corr_vals[0][1][0][2]} \n",
119     f" A'B'  {s_corr_vals[0][0][0][3]} +/- {s_corr_vals[0][1][0][3]} \n",
120     f" S-Value {s_corr_vals[1][0]} +/- {s_corr_vals[1][1]}",
121 )
122 # print(get_correlation(data_alice))

```
